

HAREESH LOTTI

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Profile Summary

Emerging **AI/ML Engineer** with hands-on experience building **RAG systems**, **vector search pipelines**, and **backend APIs** using FastAPI, LangChain, and Qdrant. Skilled in developing scalable document ingestion workflows, semantic search, and intelligent retrieval for real-world AI applications. Strong problem-solver with a focus on clean engineering, automation, and practical AI deployment.

Experience

Software Developer / AI Engineer – Dhruva Space

Feb 2026 – Present

Space Technology Solutions Company / Defence & Government Projects

- Developed and solely maintained **standalone and army-grade web applications** involving full-stack development, from architecture to production deployment.
- Built a **real-time satellite tracking UI** integrated with a backend system, enabling live monitoring and geospatial data visualization for defence use cases.
- Developed a web application with **object detection capabilities**, integrating computer vision models into a Django backend with a responsive frontend.
- Handled end-to-end ownership of features — backend APIs, frontend UI, **testing**, and **client-facing delivery** — with direct involvement in high-level meetings and requirement discussions.
- Collaborated within a **government and defence-oriented project environment**, adhering to security standards and mission-critical reliability requirements.
- Worked across the full development lifecycle including collaborative team development, code reviews, and cross-functional coordination.

AI Backend Engineer – Zudu.ai

Nov 2025 – Feb 2026

PRs At Zudu.ai [🔗](#)

Product & SaaS-Based Conversational AI Company

- Built and deployed a production-ready **RAG backend microservice** with FastAPI powering a real-time voice assistant.
- Developed a **Hybrid Search system** in Qdrant using dense + sparse scoring, improving technical document retrieval accuracy.
- Improved query relevance with **query rewriting** and optimized threshold logic.
- Designed a robust **document ingestion pipeline** with adaptive text splitting and parent-child indexing for long-context retrieval.
- Added **semantic chunking** and LLM-driven metadata extraction to enhance document understanding.
- Implemented advanced **PDF parsing** with PyMuPDF and OCR fallback for complex, image-heavy documents.
- Built a **Background Sound Mixer** for LiveKit agents, enabling real-time ambient audio overlay across 9 sound types with adjustable volume.

Projects

Ask Chatbot! – RAG & Multi-Server Agentic AI System

{Generative AI, RAG, MCP, LangGraph,

Gemini, Manim, AsyncIO, SQL, LangSmith}

RAG Repo [🔗](#) **Agent Repo** [🔗](#)

- Built a high-performance **RAG system** with semantic search and adaptive chunking using LangChain and FAISS for accurate multi-document retrieval.
- Orchestrated **Model Context Protocol (MCP)** to integrate multi-server tools (Math, Expenses, Manim), enabling autonomous animation generation and complex tool execution.
- Engineered a custom **async event-loop backend** to bridge LangGraph's async execution with synchronous Streamlit UI, supporting real-time **streaming and tool status tracking**.

- Implemented persistent session management using **aiosqlite** and ensured observability through **LangSmith** tracing and performance analysis.

Pneumonia Detection from Chest X-Ray Images using Custom CNN Architecture *{Deep Learning, Medical Imaging}* [GitHub](#) | [Project Paper](#)

- Developed a 5-layer **Custom CNN architecture** in TensorFlow/Keras for automated pneumonia detection from pediatric chest X-rays.
- Achieved **90.4% test accuracy**, with **93% precision for Pneumonia** and **87% for Normal** cases on the test dataset of 5,863 images.
- Implemented **comprehensive data augmentation** and **Batch Normalization/Dropout** techniques to counter class imbalance and overfitting.
- Proposed future enhancements including **ensemble methods**, **interpretability with Grad-CAM**, and **clinical validation**.

StudyBud — Real-Time Collaborative Study Platform *{Django 5.2, HTMX, PostgreSQL, Supabase S3, Docker, GitHub Actions, Render}*

[GitHub](#)

- Architected a **multi-model relational data layer** (Room, Message, Topic, Profile) with **ManyToMany participant tracking** and **Django Q-object search** across 3 fields, enabling full-text room discovery without raw SQL.
- Engineered **HTMX-powered partial rendering** via a dedicated `room_messages` endpoint using `select_related` to eliminate N+1 queries, delivering live message feeds without a full-page reload or WebSocket overhead.
- Built a fully automated **CI/CD pipeline** with **GitHub Actions**: Docker Buildx compiles a `linux/amd64` image on every push to `main`, pushes to Docker Hub, and triggers a **zero-downtime Render deploy** via webhook — end-to-end in a single workflow.
- Designed a **dual-environment storage architecture**: **Supabase S3-compatible object storage** with custom public-URL domain routing for production, with a single `.env` flag switching to local **SQLite + filesystem** for instant development — no code changes required.
- Automated user **Profile provisioning** using **Django post_save Signals** with a safety-net fallback to prevent profile-missing crashes across all creation paths (register, admin, shell, management commands).

Insurance Premium Prediction API *{ML, FastAPI, Docker}* [GitHub](#) | [Live API Endpoint](#)

- Engineered features (BMI, age group, lifestyle risk) and trained a **Random Forest** model to predict insurance categories, achieving **90% accuracy**.
- Built a production-ready REST API using **FastAPI** to serve the trained model and provide real-time inference.
- Implemented server-side data validation and automatic feature computation from raw inputs (e.g., age, height) using **Pydantic** models and `@computed_field` decorators.
- Designed a clear API response schema to return the predicted category, confidence score, and class probabilities.
- Containerized the complete application, model, and dependencies using **Docker** for scalable and isolated deployment.

Education

Bachelor of Technology in Computer Science and Engineering <i>Rajiv Gandhi University of Knowledge Technologies (IIIT Nuzvid), Andhra Pradesh</i>	2022–2026 <i>CGPA: 8.85</i>
Pre-University Course – PUC <i>Rajiv Gandhi University of Knowledge Technologies (IIIT Nuzvid), Andhra Pradesh</i>	2020–2022 <i>CGPA: 9.64</i>
Board of Secondary Education, Andhra Pradesh <i>Vignan Vidhya Nikethan</i>	2019–2020 <i>CGPA: 10</i>

Technical Skills

Programming: Python (advanced), C, C++, Java (basic), SQL
AI/ML Frameworks: PyTorch, TensorFlow, Scikit-learn, Keras, Transformers
Generative AI: LangChain, LangGraph, LangSmith, **Model Context Protocol (MCP)**, RAG Agents, OpenAI/Gemini/Groq APIs
NLP: Hugging Face, NLTK, spaCy, fuzzywuzzy, BeautifulSoup, SentenceTransformers
Data Analysis & Visualization: **Manim (Animation)**, Pandas, NumPy, SciPy, Matplotlib, Seaborn, Plotly
Backend Development: FastAPI, Django, Flask, Node.js, Express.js, Streamlit, REST APIs, Microservices
Frontend Development: HTML, CSS, JavaScript, React.js, **HTMX**, Responsive UI Development
Databases: PostgreSQL, **Supabase**, MySQL, MongoDB, Qdrant, FAISS, Chroma, Pinecone
MLOps & Tools: Docker, Git, **GitHub Actions**, Postman, Hugging Face Hub, CI/CD, Linux

Certifications & Achievements

Certifications	Achievements
- NPTEL – Deep Learning (<i>Silver Badge, Top 5% nationally</i>) - NPTEL – Large Language Models (LLMs) (<i>70% nationally</i>) - NPTEL – C++ Programming (<i>Top 5% nationally</i>) View All Certificates 	1st Place – Madcode Coding Competition 1st Place – Tech Hunt Competition - Organized and mentored AI/ML workshops for juniors View Achievement Proofs 